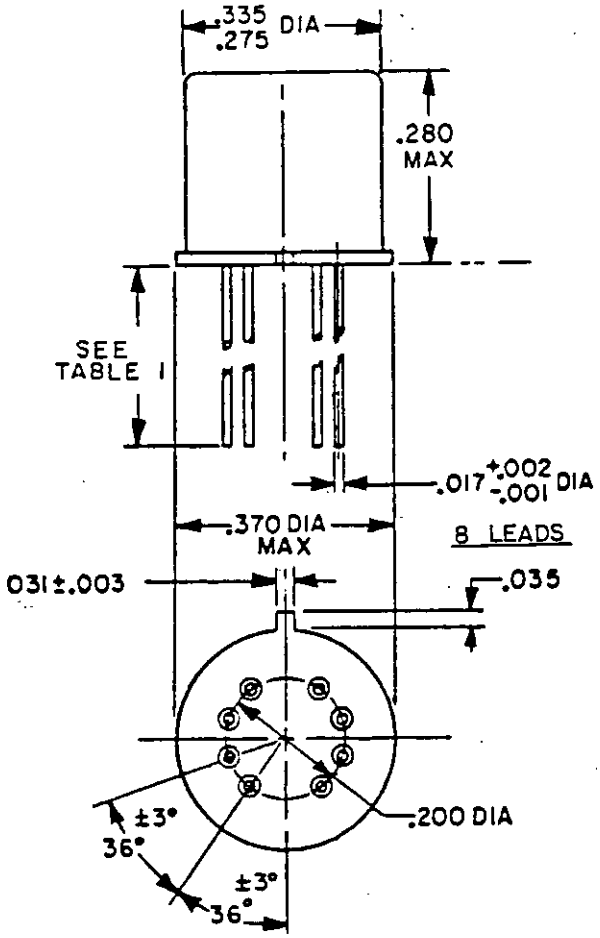
MILITARY SPECIFICATION SHEET

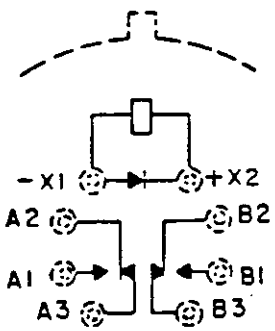
# RELAYS, ELECTROMAGNETIC, HIGH SHOCK AND HIGH VIBRATION, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 1.0 AMPERE WITH INTERNAL DIODE FOR COIL TRANSIENT SUPPRESSION

(B) This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-39016.




| Inches | mm |
|---|---|
| .001 | 0.03 |
| .002 | 0.05 |
| .003 | 0.08 |
| .017 | 0.43 |
| .031 | 0.79 |
| .035 | 0.89 |
| .200 | 5.08 |
| .275 | 6.99 |
| .280 | 7.11 |
| .335 | 8.51 |
| .370 | 9.40 |




CIRCUIT DIAGRAM
TERMINAL VIEW

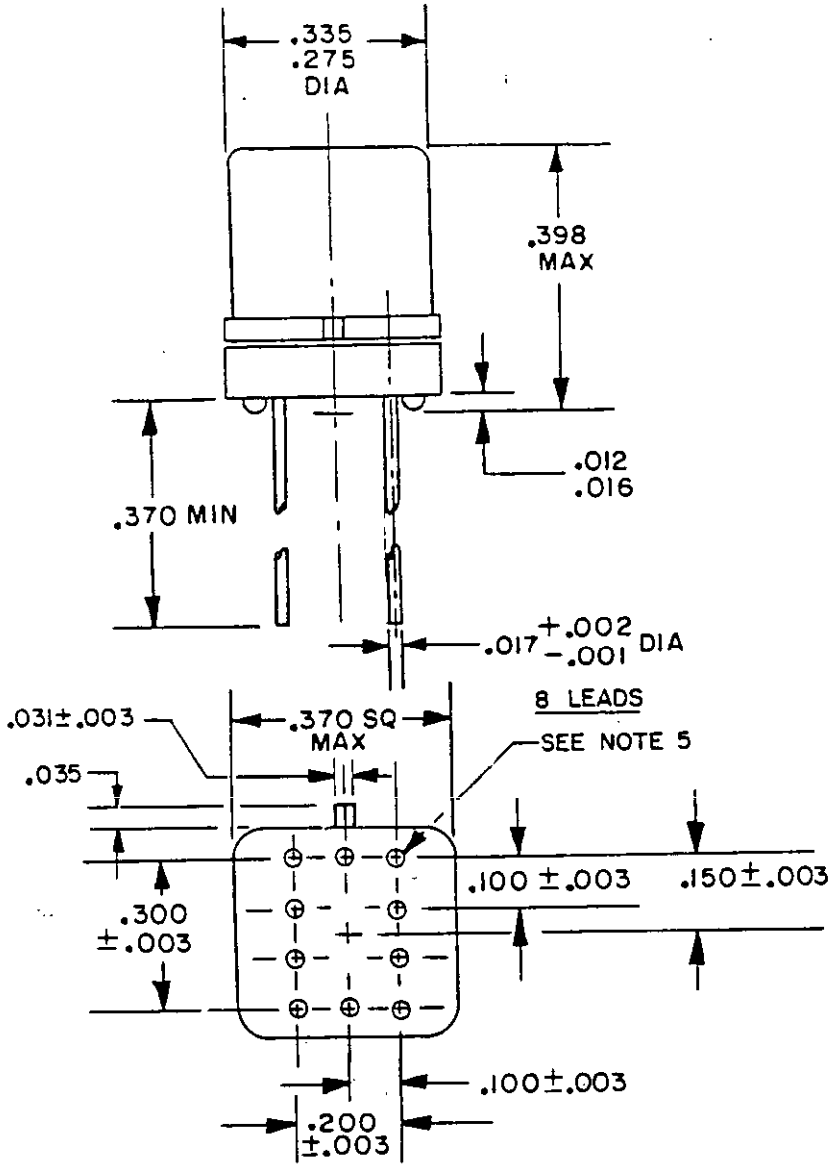
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above for reference only. Numbers do not appear on relay.
5. Coil symbol optional in accordance with MIL-STD-1285.
6. Circuit diagram shown on part is the terminal view.
7. Relays shall have a (+) sign placed on the circuit diagram as shown.

FIGURE 1. Dimensions and configuration.

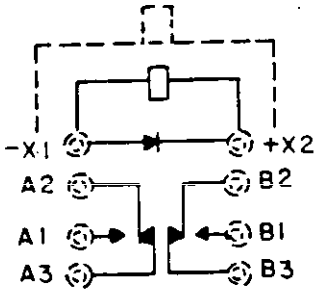
(3) denotes changes


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-21T17:52Z
Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .001 | 0.03 |
| .002 | 0.05 |
| .003 | 0.08 |
| .012 | 0.30 |
| .016 | 0.41 |
| .017 | 0.43 |
| .031 | 0.79 |
| .035 | 0.89 |
| .100 | 2.54 |
| .150 | 3.81 |
| .200 | 5.08 |
| .275 | 6.99 |
| .300 | 7.62 |
| .335 | 8.51 |
| .370 | 9.40 |
| .398 | 10.11 |




CIRCUIT DIAGRAM
TERMINAL VIEW

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader pads shall be certified to MIL-M-38527/5-03.
5. Dimensions and tolerances shown for the bottom view of the spreader pad are for the center to center locations of the holes in the spreader pad.
6. Shape optional within envelope dimension.
7. Coil symbol optional in accordance with MIL-STD-1285.
8. Terminal numbers shown above for reference only. Numbers do not appear on relay.
9. Relays shall have a (+) sign placed on the circuit diagram as shown.
10. Circuit diagram shown on part is the terminal view.

FIGURE 2. Dimensions and configuration relays supplied with spreader pad.



Source: https://assist.dla.mil -- Downloaded: 202610-21T17:52Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:

1.0 ampere at 28 V dc.

500 milliamperes at 115 V ac 400 Hz case not grounded.

250 milliamperes at 115 V ac 60 Hz case not grounded.

100 milliamperes at 115 V ac 60 and 400 Hz case grounded.

Inductive load: 0.2 ampere at 28 V dc with 0.32 henry inductance.

Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

(B) Initial: 0.10 ohm maximum (0.125 ohm maximum with spreader pad attached).

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.20 ohm maximum (0.225 ohm maximum with spreader pad attached).

Low level:

During life: 100 ohms maximum.

After life: 0.15 ohm maximum (0.175 ohm maximum with spreader pad attached).

Intermediate current:

During life: 3 ohms maximum.

After life: 0.20 ohm maximum (0.225 ohm maximum with spreader pad attached).

Contact bounce: 1.5 milliseconds maximum (applicable to failure rate level "L").

Contact stabilization time: 2.0 milliseconds maximum (applicable to failure rate levels "M", "P", and "R").

Overload (high level only): Two times rated current.

COIL DATA: See table I.

Operate time: 6.0 milliseconds maximum over temperature range with rated coil voltage.

Release time: 4.0 milliseconds maximum over temperature range from rated coil voltage.

Coil life: Applicable. (For retention of qualification testing, coil life shall be performed following vibration testing in group II of the group C inspection table of MIL-R-39016. Test time shall be 250 hours.)



Source: https://assist.dla.mil -- Downloaded: 202610-21T17:52Z
Check the source to verify that this is the current version before use.

ELECTRICAL DATA:

(B) Insulation resistance: 1/ 10,000 megohms minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

(B) Dielectric withstanding voltage: 1/

| | Sea level V rms (60 Hz) | Altitude V rms (60 Hz) |
|---|---|---|
| Between case, frame, or enclosure and all contacts both in the energized and deenergized positions | 500 | 125 All terminals to case |
| Between case, frame, or enclosure and coil(s) | 500 | |
| Between all contacts and coil(s) | 500 | |
| Between open contacts in the energized and deenergized positions | 500 | |
| Between contact poles | 500 | |

(B) DIODE CHARACTERISTICS: 2/

(B) Maximum negative transient: 1.0 volt.

(B) Peak inverse voltage: 100 volts minimum.

(B) Coil transient suppression: Applicable.

(B) Semiconductor in-process screening: Applicable, visual inspection of semiconductors shall be in accordance with MIL-STD-750, method 2073 or 2074.

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration: Applicable, except vibration level shall be 100 g's and frequency range shall be 140 to 2,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

(B) Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

(B) Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts. (Applicable to qualification and group C testing only).

Shock (specified pulse): MIL-STD-202, method 213, test condition B, except peak value shall be 150 g's for 11 milliseconds. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with MIL-STD-750, method 1041.

(B) 1/ Insulation resistance and dielectric withstanding voltage tests must always precede all other specified electrical measurements. Connect all coil terminals together to avoid damage to diodes.

(B) 2/ WARNING: Reverse polarity on coil terminals will destroy diode.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:52Z
Check the source to verify that this is the current version before use.

PHYSICAL DATA:

Terminals:

Terminal strength: Test condition A, 1 pound pull minimum.

Bend test: Test condition C, 1/2 pound load.

(B) Terminal twist test: As specified in MIL-R-39016.

Solderability: Applicable.

(B) Dimensions and configuration: See figures 1 and 2.

*Terminations*: See figures 1 and 2 and table I.

Weight: 3.1 grams (0.109 ounce) maximum (3.66 grams (0.129 ounce) maximum with spreader pad attached).

Seal: Hermetic.

Internal moisture: Applicable.

(B) Minimum marking: Military part number, "J" with the date code (example J8530), circuit diagram, manufacturer's name or source code.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per relay.

(B) Low level: 100,000 cycles plus 900,000 cycles mechanical life.

Part number: M39016/50- (dash number from table I and suffix letter designating failure rate level).

(B) TABLE I. Dash numbers and characteristics. 1/

| Dash numbers 2/ 3/ | | | Coil voltage (V dc) 4/ | | At 25°C | | | | Over temperature range | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Lead length .500 min | Lead length .187 +.040 -.010 | Spreader pads (figure 2) 3/ | Rated | Max | Coil resistance ohms ±10% | Specified pickup value (voltage) (V dc) | Specified hold value (voltage) (V dc) | Specified drop-out value (voltage) (V dc) | Specified pickup value (voltage) (V dc) | Specified hold value (voltage) (V dc) | Specified drop-out value (voltage) (V dc) |
| 001 | 007 | 013 | 5.0 | 5.8 | 50 | 3.5 | 1.4 | 0.22 | 4.6 | 2.5 | 0.14 |
| 002 | 008 | 014 | 6.0 | 8.0 | 70 | 4.2 | 2.0 | 0.27 | 5.5 | 3.2 | 0.18 |
| 003 | 009 | 015 | 9.0 | 12.0 | 155 | 6.2 | 3.0 | 0.54 | 8.2 | 4.9 | 0.35 |
| 004 | 010 | 016 | 12.0 | 16.0 | 235 | 8.3 | 4.0 | 0.63 | 11.0 | 6.5 | 0.41 |
| 005 | 011 | 017 | 18.0 | 24.0 | 610 | 12.5 | 6.0 | 0.77 | 16.5 | 10.0 | 0.59 |
| 006 | 012 | 018 | 26.5 | 32.0 | 1,130 | 18.0 | 8.0 | 1.23 | 22.0 | 13.0 | 0.89 |

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01.
Example: 001L - - - - - - 012R.

3/ Relays supplied with spreader pads (-013 through -018) shall have the pad rigidly attached.

4/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.



Source: https://assist.dla.mil -- Downloaded: 202610-21T17:52Z
Check the source to verify that this is the current version before use.

QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

(B) TABLE II. Qualification inspection and sample size. 1/

| Single submission | Group submission | |
|---|---|---|
| 18 units plus 1 open unit for level L at C = 0 2/<br>33 units plus 1 open unit for level M at C = 0 2/<br><br>Qualification inspection as applicable | M39016/50-006 | 18 units plus 1 open unit for level L at C = 0 2/<br>33 units plus 1 open unit for level M at C = 0 2/<br><br>Qualification inspection as applicable |
| | M39016/50-001<br>M39016/50-002<br>M39016/50-003<br>M39016/50-004<br>M39016/50-005 | 2 units each part number Qualification inspection, group II |

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-39016/15 may be used in addition to MIL-R-39016/50 data. Prior to performance of retention of qualification, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing will be increased as required in group V, table II, MIL-R-39016, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of qualification inspection, the relay manufacturer shall preselect the sampling plan.

Initial qualification of relays supplied with spreader pads (-013 through -018) shall be tested as specified below:

(B) Perform the following tests as specified in the qualification inspection table of MIL-R-39016, in the order shown before installation of pads; screening, visual and mechanical examination (internal), thermal shock, resistance to solvents, vibration (sinusoidal), vibration (random), shock (specified pulse), acceleration, terminal strength, magnetic interference (when specified), capacitance (when specified), coil life (applicable to continuous duty relays only), resistance to soldering heat, salt spray (corrosion), overload (applicable to high level relays only), life, terminal strength, intermediate current.

(B) After installation of pads perform the following tests as specified in the qualification inspection table of MIL-R-39016, in the order shown; insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, and visual and mechanical inspection (external).

Qualification inspection (reduced testing for previously qualified relays) for relays supplied with spreader pads (-013 through -018), 2 units of the 26.5 volt rated coil voltage (-018) shall be tested as specified below:

Before installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016 in the order shown:

For failure rate level L only: Screening.

(B) For failure rate levels M, P, and R: Vibration (sinusoidal) test duration shall be 10 minutes, vibration (random), particle impact noise detection (PIND, when specified), screening.



Source: https://assist.dla.mil -- Downloaded: 202640-21T17:52Z
Check the source to verify that this is the currentversion before use.

After installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016 in the order shown:

(B) Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, and visual and mechanical inspection (external).

Group A testing for relays supplied with spreader pads (-013 through -018), shall be tested as specified below:

(B) Before installation of pad perform subgroup 2 of group A tests.

(B) After installation of pad perform subgroups 3 and 4 of group A tests.

Qualification inspection and sample size (reduced testing) : See table III.

If the relays produced for MIL-R-39016/50 are similar in construction and design *except for the coil assembly to the relays produced for* MIL-R-39016/15, then reduced testing for qualification may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/15. For reduced testing, see table III.

(B) TABLE III. Qualification inspection (reduced testing).

| Examination or test |
|---|
| 2 units tested as specified below: |
| 26.5 V dc coil voltage |
| Group II of qualification inspection table |
| Shock |
| Vibration (as specified for qualification) |
| Life |
| All other coil voltages |
| Group II of qualification inspection table |

## CONCLUDING MATERIAL

Custodians:
(B) Army - ER
(B) Navy - EC
(B) Air Force - 85

Review activities:
(B) Navy - AS
(B) Air Force - 99
DLA - ES
NSA - NS

User activities:
(B) Army - AR
(B) Navy - MC, OS, SH
(B) Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5945-0757-42)



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:52Z
Check the source to verify that this is the current version before use.